第四章 对称性与守恒量

1、

$\frac{d}{dt}\langle\psi|\hat{F}|\psi\rangle = \frac{\partial}{\partial t}\langle\psi|\hat{F}|\psi\rangle + \langle\psi|\frac{\partial\hat{F}}{\partial t}|\psi\rangle + \langle\psi|\hat{F}\frac{\partial}{\partial t}|\psi\rangle$

$\hat{F} = \hat{F}^{\dagger}$ ，$\hat{F}^{\dagger} = \hat{F}$

$\frac{\partial}{\partial t}\langle\psi| = -\frac{1}{i\hbar}\langle\psi|\hat{H}$ ，$\frac{\partial}{\partial t}|\psi\rangle = \frac{1}{i\hbar}\hat{H}|\psi\rangle$

$= \langle\psi|\hat{F}\hat{H} - \hat{H}\hat{F} + \frac{\partial\hat{F}}{\partial t}|\psi\rangle$

$= \langle\psi|\frac{[\hat{F},\hat{H}]}{i\hbar} + \frac{\partial\hat{F}}{\partial t}|\psi\rangle$

若 $[\hat{F},\hat{H}] = 0$ 且 $\hat{F}$ 不含时（$\frac{\partial\hat{F}}{\partial t} = 0$）

则 $\frac{d}{dt}\langle\psi|\hat{F}|\psi\rangle = 0$，$\hat{F}$ 为守恒量（期待值不随时间变化）

2、力学量算符

守恒量对应的厄米算符的本征值均为实数

Ehrenfest定理：

$\frac{d}{dt}\langle\vec{r}\rangle = \frac{1}{i\hbar}\langle[\vec{r},\frac{\vec{p}^2}{2m} + \hat{V}(r)]\rangle = \frac{1}{2mi\hbar}\langle[\vec{r},\vec{p}^2]\rangle = \frac{\langle\vec{p}\rangle}{m}$

$\frac{d}{dt}\langle\vec{p}\rangle = \frac{1}{i\hbar}\langle[\vec{p},\frac{\vec{p}^2}{2m} + \hat{V}(r)]\rangle = \frac{1}{i\hbar}\langle[\vec{p},\hat{V}(r)]\rangle = -\nabla V(r)$

$\Rightarrow \frac{d\vec{r}}{dt} = \frac{\vec{p}}{m}$，$\frac{d\vec{p}}{dt} = -\nabla V$

能量-时间不确定关系：

$\Delta\hat{A}\cdot\Delta\hat{A} \geq \frac{1}{4}|[\hat{A},\hat{H}]|^2 = \frac{\hbar^2}{4}|\frac{d\hat{A}}{dt}|^2$

令 $\Delta t = \frac{\Delta\hat{A}}{|\frac{d\hat{A}}{dt}|}$，$\Delta\hat{A}\cdot\Delta t \geq \frac{\hbar}{2}$

$t \geq \Delta t$ 才能看出显著变化（正如子弹落）

§ 4.2 对称性与守恒量的关系

存在正变换 $\hat{Q}$ 下

$|\psi\rangle \to |\psi'\rangle = \hat{Q}|\psi\rangle$

对称变换：$\langle a|b\rangle = \langle a'|b'\rangle \quad \forall |a\rangle, |b\rangle$ 成立.

Wigner 定理：对称变换 = 幺正变换 / 反幺正变换

$\langle\psi|\hat{A}|\psi\rangle \to \langle\psi'|\hat{A}|\psi'\rangle = \langle\psi|\hat{Q}^\dagger\hat{A}\hat{Q}|\psi\rangle \quad |\psi'\rangle = \hat{Q}|\psi\rangle$

$\Rightarrow$ 在幺算符的变换 $\hat{A} \to \hat{A}' = \hat{Q}^\dagger\hat{A}\hat{Q}$

系统对称 $\hat{Q}$ 变换下有对称性：Schrodinger 方程不变.

$\hat{Q}^\dagger\hat{H}\hat{Q} = \hat{H} \Rightarrow [\hat{Q}, \hat{H}] = 0$

连续变换 $\hat{Q}$：

幺正：$\hat{Q}^\dagger\hat{Q} = \hat{Q}\hat{Q}^\dagger = 1$

无穷小变换：$\hat{Q}(\eta) = e^{i\eta\hat{F}} = 1 + i\eta\hat{F}$

$\hat{Q}^\dagger\hat{Q} = (1 - i\eta\hat{F}^\dagger)(1 + i\eta\hat{F}) = 1 + i\eta(\hat{F} - \hat{F}^\dagger) = 1$

$\hat{F} = \hat{F}^\dagger$，$\hat{F}$ 是厄米算符. $\hat{F}$：连续变换的生成算符.

对称性：$[\hat{Q}, \hat{H}] = [e^{i\eta\hat{F}}, \hat{H}] = 0$ 请完成.

$\Rightarrow [\hat{F}, \hat{H}] = 0$

2. 平移算符 $\hat{T}$

$x \to x' = x + \delta x$

$\hat{T}(\delta x)|x\rangle = |x + \delta x\rangle$

$|\psi'\rangle = \hat{T}|\psi\rangle = \int \hat{T}|x\rangle\langle x|\psi\rangle\, dx$

$= \int dx\, |x + \delta x\rangle\langle x|\psi\rangle = \int dx' |x'\rangle\langle x' - \delta x|\psi\rangle$

$= \int dx' |x'\rangle \psi(x - \delta x)$

$\Rightarrow \psi'(x) = \langle x'|\psi'\rangle = \psi(x - \delta x)$

具体形式：$\langle x|\hat{T}(\delta x)|\psi\rangle$

$= \langle x|1 + i\delta x\hat{F}|\psi\rangle = \langle x|\psi\rangle + i\delta x\langle x|\hat{F}|\psi\rangle$

$\psi(x - \delta x)$ 展开 $= \psi(x) - \delta x\frac{d}{dx}\psi(x)$

$\Rightarrow \hat{F} = -\hat{p}_x$ 平移对称性：$[\hat{p}_x, \hat{H}] = 0$ 动量守恒.

$$\hat{T}(x) = \lim_{N\to\infty} \int e^{-\frac{ix}{N}\frac{\hat{p}}{\hbar}} \int e^{i\delta x \cdot \frac{\hat{p}}{\hbar}} = e^{-\frac{i\hat{p}x}{\hbar}}$$

任意算符的变换：

$$\hat{x}' = \hat{T}^{+}\hat{x}\hat{T} = e^{\frac{i\hat{p}x}{\hbar}}\hat{x}\,e^{-\frac{i\hat{p}x}{\hbar}} = \hat{x} + a$$

Baker - Hausdorff 定理：

$$f(\lambda) = e^{\lambda\hat{A}}\hat{B}\,e^{-\lambda\hat{A}}$$

$$f'(\lambda) = e^{\lambda\hat{A}}\hat{A}\hat{B}\,e^{-\lambda\hat{A}} - e^{\lambda\hat{A}}\hat{B}\hat{A}\,e^{-\lambda\hat{A}}$$

$$= e^{\lambda\hat{A}}[\hat{A},\hat{B}]\,e^{-\lambda\hat{A}}$$

$$f''(\lambda) = e^{\lambda\hat{A}}([\hat{A},[\hat{A},\hat{B}]] - [\hat{A},\hat{B}]\hat{B})\,e^{-\lambda\hat{A}}$$

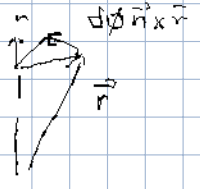
$$= e^{\lambda\hat{A}}[\hat{A},[\hat{A},[\hat{A},\hat{B}]]]\,e^{-\lambda\hat{A}}$$

$$f^{(n)}(\lambda) = e^{\lambda\hat{A}}[\hat{A},\cdots[\hat{A},[\hat{A},\hat{B}]]\cdots]\,e^{-\lambda\hat{A}}$$

$$\Rightarrow f(\lambda) = f(0) + \sum_n \frac{\lambda^n}{n!} f^{(n)}(\lambda)$$

$$= \sum_n \frac{\lambda^n}{n!} C_n \qquad C_0 = \hat{B},\ C_1 = [\hat{A},\hat{B}],\ C_2 = [A,C_1],\cdots$$

3. 旋转算符 $\hat{R}_n$



$$\vec{r}' = \vec{r} + \delta\phi\,\vec{n}\times\vec{r}$$

$$\hat{R}_n(\delta\phi),\quad \hat{R}_n(\delta\phi)|\vec{r}\rangle = |\vec{r} - \delta\phi\,\vec{n}\times\vec{r}\rangle$$

$$\psi'(r) = \langle r|\hat{R}_n(\delta\phi)|\psi\rangle = \psi(\vec{r} - \delta\phi\,\vec{n}\times\vec{r})$$

$$\langle r|\hat{R}_n(\delta\phi)|\psi\rangle = \psi(r) - \delta\phi\,(\vec{n}\times\vec{r})\cdot\nabla\psi(r)$$ 泰勒展开

$$= \langle r|\psi\rangle - \frac{i}{\hbar}\delta\phi\,\langle r|(\vec{n}\times\vec{r})\cdot\hat{p}|\psi\rangle \qquad \hat{p} = -i\hbar\nabla$$

$$= \langle r|\psi\rangle - \frac{i}{\hbar}\delta\phi\,\langle r|(\vec{r}\times\hat{p})\cdot\vec{n}|\psi\rangle$$

$$= \langle r|1 - \frac{i}{\hbar}(\vec{L}\cdot\vec{n})\delta\phi|\psi\rangle$$

$$\Rightarrow \hat{R}_n(\delta\phi) = 1 - \frac{i}{\hbar}(\vec{L}\cdot\vec{n})\delta\phi = e^{-\frac{i(\vec{L}\cdot\vec{n})\delta\phi}{\hbar}}$$

$$\hat{R}_n(\phi) = \prod \hat{R}_n(\delta\phi) = e^{-\frac{i}{\hbar}\vec{L}\cdot\vec{n}\,\phi}$$

旋转对称：$[\vec{L}\cdot\vec{n},\hat{H}] = 0$ 角动量守恒

离散对称性：

空间反演：$\hat{\Pi}: \vec{r} \to -\vec{r}$

$\hat{\Pi}|\vec{r}\rangle = |-\vec{r}\rangle$，$\hat{\Pi}|\vec{p}\rangle = |-\vec{p}\rangle$

坐标、动量算符的变换：

$\hat{\Pi}\hat{r}\hat{\Pi}^{\dagger}\hat{\Pi}|\vec{r}\rangle = \hat{\Pi}\hat{r}|\vec{r}\rangle = \hat{\Pi}\vec{r}|\vec{r}\rangle$

$= \vec{r}\hat{\Pi}|\vec{r}\rangle = \vec{r}|-\vec{r}\rangle$

$= -\vec{r}|-\vec{r}\rangle = -\vec{r}\hat{\Pi}|\vec{r}\rangle$

$\Rightarrow \hat{\Pi}^{\dagger}\hat{r}\hat{\Pi} = -\vec{r}$ 同理，$\hat{\Pi}^{\dagger}\hat{p}\hat{\Pi} = -\vec{p}$

$\hat{\Pi}^{\dagger}\{\hat{x}\cdot\hat{p}\}\hat{\Pi} = \hat{\Pi}^{\dagger}(\hat{x}\hat{p} - \hat{p}\hat{x})\hat{\Pi}$

$= \hat{\Pi}^{\dagger}\hat{x}\hat{\Pi}\hat{\Pi}^{\dagger}\hat{p}\hat{\Pi} - \hat{\Pi}^{\dagger}\hat{p}\hat{\Pi}\hat{\Pi}^{\dagger}\hat{x}\hat{\Pi}$

$= \hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar$

* $\hat{\Pi}^{\dagger}\hat{\Pi} = 1$ ⇒ $\hat{\Pi}$ 是幺正算符。

$\hat{\Pi}|\psi\rangle = \hat{\Pi}\int d\vec{r}^3 |\vec{r}\rangle\langle\vec{r}|\psi\rangle = \int d\vec{r}^3 |-\vec{r}\rangle\langle\vec{r}|\psi\rangle$

$\vec{r} \to -\vec{r}$ $= -\int_{+\infty}^{-\infty} d\vec{r}^3 |\vec{r}\rangle\langle-\vec{r}|\psi\rangle = -\int_{+\infty}^{-\infty} d\vec{r}^3 |\vec{r}\rangle\psi(-\vec{r})$

$= \int_{-\infty}^{+\infty} d\vec{r}^3 |\vec{r}\rangle\psi(-\vec{r}) = |\psi(-\vec{r})\rangle$

$\psi'(\vec{r}) = \langle\vec{r}|\hat{\Pi}|\psi\rangle = \psi(-\vec{r})$

$\hat{\Pi}^2|\vec{r}\rangle = |\vec{r}\rangle$ $\hat{\Pi}^2 = 1$ ⇒ $\hat{\Pi}$ 是幺正算符

若能本解：$\hat{\Pi}|\psi\rangle = \lambda|\psi\rangle$

$\hat{\Pi}^2|\psi\rangle = \lambda^2|\psi\rangle = |\psi\rangle$ ⇒ $\lambda = \pm 1$ $\lambda = 1$，偶宇称。

$\lambda = -1$，奇宇称。

系统有空间反演对称性：

$[\hat{\Pi}, \hat{H}] = 0$ 宇称守恒。

例：一维谐振子。

$\hat{\Pi}|n\rangle = \alpha_n|n\rangle$，$\alpha_n = \pm 1$ $\hat{\Pi}\hat{a}^{\dagger}\hat{\Pi} = -\hat{a}^{\dagger}$

$\hat{\Pi}|n\rangle = \hat{\Pi}\frac{1}{\sqrt{n}}\hat{a}^{\dagger}|n-1\rangle = -\frac{1}{\sqrt{n}}\hat{a}^{\dagger}\hat{\Pi}|n-1\rangle$

$= -\frac{1}{\sqrt{n}}\hat{a}^{\dagger}\alpha_{n-1}|n-1\rangle = -\alpha_{n-1}|n\rangle$

⇒ $\alpha_n = -\alpha_{n-1}$，$\alpha_n = (-1)^n\alpha_0$

其态：偶宇称 $\alpha_n = (-1)^n$，$\hat{P}|n\rangle = (-1)^n|n\rangle$

§4.3 时间反演对称性

$t \to -t$

$\hat{\Theta}|\vec{r}\rangle = |\vec{r}\rangle$，$\hat{\Theta}|\vec{p}\rangle = |-\vec{p}\rangle$

反幺正算符：$\hat{\Theta}^\dagger\hat{\Theta} = 1$

$|\alpha'\rangle = \hat{\Theta}|\alpha\rangle$，$|\beta'\rangle = \hat{\Theta}|\beta\rangle$ $\langle\alpha'|\beta'\rangle = \langle\alpha|\beta\rangle^*$

反线性 $\hat{\Theta}(c_\alpha|\alpha\rangle + c_\beta|\beta\rangle) = c_\alpha^*\hat{\Theta}|\alpha\rangle + c_\beta^*\hat{\Theta}|\beta\rangle$

$\hat{\Theta}\hat{r}\hat{\Theta} = \hat{r}$，$\hat{\Theta}\hat{p}\hat{\Theta} = -\hat{p}$

表示：

$|\psi'\rangle = \hat{\Theta}|\psi\rangle = \sum_n \hat{\Theta}|n\rangle\langle n|\psi\rangle^*$

$= \sum_n |n'\rangle\langle\psi|n\rangle$

$\hat{\Theta} = \hat{U}\hat{K}$ $\hat{U}$ 为幺正算符，负责对其实变换。

Wigner定理 $\hat{K}$ 负责对展开系数取复共轭。（反线性）

$\hat{\Theta}^{-1} = \hat{K}^{-1}\hat{U}^{-1} = \hat{K}\hat{U}^\dagger$

对坐标算符的变换：

$\hat{\Theta}|\psi\rangle = \hat{\Theta}\int d^3r\,|r\rangle\langle r|\psi\rangle = \int d^3r\,|r\rangle\langle r|\psi\rangle^*$

$\hookrightarrow \psi'(r) = \langle r|\hat{\Theta}|\psi\rangle = \psi^*(r)$

$\hat{p}$ 对算符的变换：$\hat{K}^2 = 1$，$\hat{K} = \hat{K}^{-1}$

$\hat{K}\hat{A}\hat{K}^{-1}|\psi\rangle = \hat{K}\hat{A}\hat{K}|\psi\rangle = \hat{K}\hat{A}|\psi^*\rangle$（$\hat{K}|\psi\rangle = |\psi^*\rangle$）

$= (\hat{A}|\psi^*\rangle)^* = \hat{A}^*|\psi\rangle$

$\to \hat{K}\hat{A}\hat{K}^{-1} = \hat{A}^*$

$\hat{\Theta}^2 = \hat{U}\hat{K}\hat{U}\hat{K}$

$= \hat{U}\hat{K}\hat{U}\hat{K}^{-1} = \hat{U}\hat{U}^* = \alpha$ α 为模长为1，为实数。

Schur引理：若一个有限维不可约表示空间 此上有线性算符 $\hat{A}$ 满足 $\hat{A}$ 和所有表示中的变换 $D(g)$ 对易，即 $\hat{A} \propto I$

$\hookrightarrow \hat{U} = \alpha\hat{U}^T$，$\hat{U}^T = \alpha\hat{U}$ $\to \hat{U} = \alpha^2\hat{U}$，$\alpha = \pm 1$

例：一个具有时间反演对称性的系统，如果能级 E_n 不简并，则相应本征函数可以表示为实函数。

$\because \hat{\Theta}\hat{H}|n\rangle = \hat{H}\hat{\Theta}|n\rangle = E_n\hat{\Theta}|n\rangle$，$\hat{H}|n\rangle = E_n|n\rangle$

$\therefore \hat{\Theta}|n\rangle = k_n|n\rangle$

$\langle r|\hat{\Theta}|n\rangle = \langle r|n\rangle^* \Rightarrow \langle r|n\rangle = e^{i\delta}\langle r|n\rangle^*$

所以 $\langle r|n\rangle$ 为实数

§4.4. 轨道角动量本征态的性质

本征态：$|l,m\rangle$

旋转：$\hat{R}_{\vec{n}}(2\pi)|l,m\rangle = e^{-i2m\pi}|l,m\rangle$

守恒：在空间反演下 $\hat{\pi}^\dagger \hat{L} \hat{\pi} = \hat{\pi}^\dagger \hat{r} \hat{\pi} \times \hat{\pi}^\dagger \hat{p} \hat{\pi} = (-\hat{r}) \times (-\hat{p}) = \hat{L}$

$\Rightarrow [\hat{\pi}, \hat{L}] = 0$，$[\hat{\pi}, \hat{L}^2] = 0$

具有本征态 $|l,m\rangle$ 有确定的宇称

$[\hat{L}_\pm, \hat{L}_\pm] = 0 \Rightarrow \hat{L}_\pm$ 不改变 $|l,m\rangle$ 的宇称。

$\Rightarrow |l,m\rangle$ 的宇称与 m 无关，只与 l 有关。

具体关系：

$[\hat{r}, \hat{L}^2] = -2i\hbar\, \hat{r} \times \hat{L} - 2\hbar^2 \hat{r}$

证：$[\hat{r}_i, \hat{L}_j] = \varepsilon_{jkm} [\hat{r}_i, \hat{r}_k \hat{p}_m] = \varepsilon_{jkm} \hat{r}_k [\hat{r}_i, \hat{p}_m] = i\hbar \varepsilon_{ijk} \hat{r}_k$

$[\hat{r}_i, \hat{L}_j \hat{L}_j] = [\hat{r}_i, \hat{L}_j] \hat{L}_j + \hat{L}_j [\hat{r}_i, \hat{L}_j]$

$= i\hbar \varepsilon_{ijk} (\hat{r}_k \hat{L}_j + \hat{L}_j \hat{r}_k)$

$= -i\hbar \varepsilon_{ikj} 2 \hat{r}_k \hat{L}_j + i\hbar \varepsilon_{ijk} (-\hat{r}_k \hat{L}_j + \hat{L}_j \hat{r}_k)$

$= -2i\hbar (\hat{r} \times \hat{L})_i - i\hbar \varepsilon_{ijk} [\hat{r}_k, \hat{L}_j]$

$= -2i\hbar (\hat{r} \times \hat{L})_i - (i\hbar)^2 \varepsilon_{ijk} \varepsilon_{kji} \hat{r}_i$

$= -2i\hbar (\hat{r} \times \hat{L})_i - \hbar^2 \hat{r}_i$ □

或：$[\hat{r}, \hat{L}^2] = [\hat{r}, \hat{L}] \hat{L} + \hat{L} [\hat{r}, \hat{L}]$

$[\hat{r}, \hat{L}] = i\hbar (\hat{r} \times \hat{1})$

$\hookrightarrow [\hat{r}, \hat{L}^2] = i\hbar [(\hat{r} \times \hat{1}) \cdot \hat{L} + \hat{L} \cdot (\hat{r} \times \hat{1})]$

$= i\hbar (\hat{L} \times \hat{r} - \hat{r} \times \hat{L})$

$= i\hbar \cdot -2 (\hat{r} \times \hat{L})$

$[[\hat{r}, \hat{L}^2], \hat{L}^2] = [-2i\hbar\, \hat{r} \times \hat{L}, \hat{L}^2] - 2\hbar^2 [\hat{r}, \hat{L}^2]$

$= -2i\hbar [\hat{r}, \hat{L}^2] \times \hat{L} - 2\hbar^2 \hat{r} (i\hbar) = 4\hbar^2 \hat{r} \times \hat{L} - 2i\hbar^3$

$[\hat{A}, (\vec{B} \times \vec{C})_i] = [\hat{A}, \sum_{jk} \varepsilon_{ijk} B_j C_k]$

$= \sum_{jk} \varepsilon_{ijk} \{ [\hat{A}, B_j] \hat{C}_k + B_j [\hat{A}, \hat{C}_k] \}$

$= \{ [\hat{A}, \hat{B}] \times \hat{C} + \hat{B} \times [\hat{A}, \hat{C}] \}_i$

$$\langle l', m' | \{ \vec{L}^2, [\vec{L}^2, \vec{L}^2] \} | l, m \rangle$$

① $= 2\hbar^2 \langle l', m' | (\vec{L}^2 \vec{L}^2 + \vec{L}^2 \vec{L}^2) | l, m \rangle$

$= 2\hbar^2 \left[\langle l', m' | \vec{L}^2 \vec{L}^2 | l, m \rangle + \langle l', m' | \vec{L}^2 \vec{L}^2 | l, m \rangle \right]$

$= 2\hbar^4 \left[l(l+1) + (l+1)l \right] \langle l', m' | \vec{L} | l, m \rangle$

② $= \langle l', m' | [\vec{L}^2, [\vec{L}^2, \vec{L}]] | l, m \rangle$

$= \hbar^2 \left\{ (l+2)(l+1) - (l+1)l \right\} \langle l', m' | [\vec{L}^2, \vec{L}] | l, m \rangle$

$= \hbar^2 \left\{ (l+2)(l+1) - (l+1)l \right\} \langle l', m' | \vec{L}^2 \vec{L} - \vec{L} \vec{L}^2 | l, m \rangle$

$= \hbar^4 \left\{ (l+2)(l+1) - (l+1)l \right\}^2 \langle l', m' | \vec{L} | l, m \rangle$

$\langle l', m' | \vec{L} | l, m \rangle \neq 0$

$\Leftarrow \left\{ (l+2)(l+1) - (l+1)l \right\}^2 = 2 \left\{ (l+2)(l+1) + (l+1)l \right\}$

$0 = (l+2)(l+1) - (l+1)l \quad 2 - (l+2)(l+1) 2 - 2(l+1)^2 + 2(l+2)^2$

$\left\{ 1 - 2(l+1) \right\} \left\{ 1 - 2(l+1) \right\} = 0$

解答 $l' = l - 1, l, l + 1$

作业：

1. 证 $[\hat{T}(\vec{a}), \hat{x}]|\vec{r}\rangle = \hat{T}(\vec{a})\hat{x}|\vec{r}\rangle - \hat{x}\hat{T}(\vec{a})|\vec{r}\rangle$

$= \hat{T}(\vec{a})|-\vec{r}\rangle - \hat{x}|\vec{r}+\vec{a}\rangle = |-\vec{r}+\vec{a}\rangle - |-\vec{r}-\vec{a}\rangle$

$[\hat{R}_n(\varphi), \hat{x}]|\vec{r}\rangle = \hat{R}_n(\varphi)|-\vec{r}\rangle - \hat{x}|R\vec{r}\rangle$

$= |R(-\vec{r})\rangle - |-R\vec{r}\rangle = 0$

证 $[\hat{H}, \hat{P}] = 0$, $[\hat{H}, \hat{\theta}] = 0$

$|p\rangle$ 为 $\hat{H}, \hat{P}$ 的共同本征态，设本征值为 H_p：$\hat{H}|p\rangle = H_p|p\rangle$

$\hat{H}|-p\rangle = H_p|-p\rangle$

$= \hat{\theta}^{-1}\hat{H}\hat{\theta}|-p\rangle = \hat{\theta}^{-1}\hat{H}|p\rangle = H_p\hat{\theta}^{-1}|p\rangle = H_p|-p\rangle$

得 $H_p = H_{-p}$，H 是 p 的偶函数 → 开中只有 p 的偶次项？

反：$[\hat{p}, \hat{H}] = 0$ → 开不依赖于 $\vec{r}$

$\hat{P}\hat{x}|p\rangle = -p|-p\rangle$，$\hat{x}\hat{P}|p\rangle = p|-p\rangle$

⇒ 为 $|p\rangle$ 的共同本征态。

2. 证 $\hat{T} = e^{-\frac{i\vec{p}\cdot\vec{a}}{\hbar}}$

$[\hat{p}, \hat{T}] = 0$

$[\hat{r}, \hat{T}]|\vec{r}\rangle = (\hat{r}\hat{T}|\vec{r}\rangle - \hat{T}\hat{r}|\vec{r}\rangle) = \vec{r}|\vec{r}+\vec{a}\rangle - r\hat{T}|\vec{r}\rangle$

$= \vec{a}|\vec{r}+\vec{a}\rangle = \vec{a}\hat{T}|\vec{r}\rangle$

$[\hat{H}, \hat{T}] = \vec{e}_z \cdot \{[\alpha(\hat{L}\times\hat{p}), \hat{T}] + \beta[\hat{p}, \hat{T}]\}$

$= \vec{e}_z \cdot \alpha[\hat{L}, \hat{T}]\times\hat{p} = \vec{e}_z \cdot \alpha(\vec{r}, \hat{T}]\times\hat{p})\times\hat{p}$

$= \alpha\, \vec{e}_z \cdot \vec{a}\,\vec{T}\times\hat{p})\times\hat{p}$

→ $\alpha = 0$ 或 $\vec{e}_z \cdot \vec{a} = 0$

证 $[\hat{L}, \hat{L}_z] = [\hat{L}_x, \hat{L}_z] + [\hat{L}_y, \hat{L}_z] = i\hbar(\hat{L}_x - \hat{L}_y)$

$[\hat{p}, \hat{L}_z] = [\hat{p}, x\hat{p}_y - y\hat{p}_x] = -i\hbar(\hat{p}_y - \hat{p}_x) = i\hbar(\hat{p}_x - \hat{p}_y)$

$[\hat{H}, \hat{L}_z] = \vec{e}_z \cdot \{\alpha[\hat{L}\times|\vec{p}, \hat{L}_z] + [\hat{L}, \hat{L}_z]\times\hat{p}) + \beta[|\vec{p}, \hat{L}_z]\}$

$= i\hbar\ \alpha\left[-(L_x p_y + L_y p_x) + L_x p_y + L_y p_x\right]$

$= 0$

(3) $\hat{\Pi}\hat{\Pi} = \vec{e}_z \cdot \left[\alpha\,\vec{\Pi}(\vec{L}\times\vec{p})\vec{\Pi} + \beta\,\vec{\Pi}\vec{p}\vec{\Pi}\right]$

$= \vec{e}_z \cdot \left[\alpha\,(\vec{\Pi}\vec{L}\vec{\Pi}\times\vec{\Pi}\vec{p}\vec{\Pi}) - \beta\vec{p}\right]$

$= \vec{e}_z \cdot \left[\alpha\,(\vec{\Pi}\vec{r}\vec{\Pi}\times\vec{\Pi}\vec{p}\vec{\Pi})\times\vec{\Pi}\vec{p}\vec{\Pi} - \beta\vec{p}\right]$

$= -\beta\vec{p} \quad \to \alpha, \beta = 0$

(4) $\vec{\theta}\hat{\Pi}\vec{\theta} = \vec{e}_z \cdot (\alpha\,\vec{\theta}\vec{L}\vec{\theta}\times\vec{\theta}\vec{p}\vec{\theta} + \beta\,\vec{\theta}\vec{p}\vec{\theta})$

$= \vec{e}_z \cdot \left[\alpha\,(\vec{\theta}\vec{r}\vec{\theta}\times\vec{\theta}\vec{p}\vec{\theta})\times\vec{\theta}\vec{p}\vec{\theta} + \beta\,\vec{\theta}\vec{p}\vec{\theta}\right]$

$= \vec{e}_z \cdot \left[\alpha(\vec{L}\times\vec{p}) - \beta\vec{p}\right]$

$\to$ α 任意，$\beta = 0$

$\hat{H}^+ = \hat{H}$

(5) $\hat{H}^+ = (\alpha(\hat{L}_x\hat{p}_y - \hat{L}_y\hat{p}_x) + \beta\hat{p}_z)^*$

$= \alpha^*(\hat{p}_y\hat{L}_x - \hat{p}_x\hat{L}_y) + \beta^*\hat{p}_z$ $\quad$ $[\hat{L}_x, \hat{p}_y] = [\hat{L}_y, \hat{p}_x]$

$= \alpha^*(\hat{L}_x\hat{p}_y - \hat{L}_y\hat{p}_x) + (\beta^* - 2i\hbar\alpha)\hat{p}_z$

α 为实数，$\mathrm{Im}\,\beta = \hbar\alpha$

$[\hat{H}, \hat{p}] = \alpha\,[(\hat{L}_x\hat{p}_y - \hat{L}_y\hat{p}_x), \hat{p}] \neq 0$

$\to$ $\alpha = 0$ 时才有平移对称性。

$[\hat{H}, \hat{L}_z] = 0 \to$ 系统具有绕 z 轴的任意角对称性。

$\hat{\Sigma}^+\hat{H}\hat{\Sigma} = -\hat{H} \neq \hat{H}$ $\quad (\vec{p} \to -\vec{p})$

$\hat{\Theta}^+\hat{H}\hat{\Theta} = \alpha^*(L_x p_y - L_y p_x) - \beta^* p_z$ $\quad$ $\vec{p} \to -\vec{p}$，$\vec{L} \to -\vec{L}$，$i \to -i$

$\therefore \beta = -i\hbar\alpha$.

β が純虚数、$\beta^* = -\beta$

$= \alpha^*(L_x p_y - L_y p_x) + \beta p_z$